NET//ETHER

Field Network Management Tool

User Guide

Application version 6.2.0 · Document version 3.0

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1. What it is

NET//ETHER is a compact, always-on-top Windows tool for low-voltage field technicians. It does four things fast: switches the laptop's wired adapter between IP configurations, monitors connectivity, scans a subnet to see what's there, and remembers what was found at each site so the next visit starts with knowledge instead of guesswork.

It runs elevated because adapter changes go through Windows' own netsh. On a company-managed laptop that elevation is granted silently by policy; on an unmanaged PC Windows asks once at launch. There are no prompts per operation.

> Everything the app knows lives on your laptop under %APPDATA%\net-ether. Nothing is sent anywhere, except an optional vendor lookup for unknown MAC addresses that you can turn off.

2. The window

Control	What it does
NET//ETHER logo	Drag the window from anywhere on the title bar.
Version chip (e.g. v6.2.0)	Opens DIAGNOSTICS. Turns amber if the app is not running elevated.
◑	Toggle 50% transparency so you can read what's behind the HUD.
▣	Colour theme.
?	Quick guide for the tab you're on. FULL MANUAL → opens this document.
–	Hide to the system tray. Click the tray icon to bring it back.
×	Quit. If you changed any adapter this session you're offered a one-click restore first.

The tray icon pulses green normally, amber when the HUD is dimmed, and flashes red while any PING host is failing — even with the HUD hidden.

3. ETHER — adapter configuration

Everything on this tab acts on the adapter chosen in the ADAPTER dropdown. Wireless, VPN and virtual adapters are filtered out; only wired Ethernet is listed, each with its connection state and a DHCP or STATIC badge.

Presets

- **LIVE** always mirrors the adapter's current settings and is never saved.
- Three named slots are yours. Fill in the fields and press **SAVE TO PRESET**. Double-click a slot name to rename it.

Known defaults

Tiles for common device brands fill in that device's factory subnet and put you one address above it. One exception:

> AXIS — current Axis cameras ship DHCP-only with no factory address. Without a DHCP server they fall back to a random 169.254.x.x address. The AXIS tile puts your laptop on 169.254.1.1 / 255.255.0.0, which reaches

> any of them; read the camera's actual address from its label or the Axis discovery tool, then open it in the browser.

Buttons

Button	Action
APPLY CONFIG	Writes IP, mask, gateway and DNS to the adapter, then re-reads it to confirm the change took. The previous configuration is snapshotted first.
DHCP	Returns the adapter to DHCP.
MTU	Resets MTU to 1500 if it isn't already. Fixes "can ping, can't load web pages".
CMD	Shows the exact netsh commands without running them, with a COPY button for an admin prompt.
REVERT	Restores the configuration from before your last APPLY.

A gateway is optional — leave it blank for direct device connections. 169.254 addresses are allowed; a note reminds you that 255.255.0.0 is usually the mask you want there.

Multi-IP

Adds a second IP to the adapter so you can reach a device on another subnet without losing your primary address.

- Enter the IP and mask, press **ADD MULTI-IP**. The app confirms the address is really present before reporting success.
- If the adapter is on DHCP, adding a multi-IP converts it to static (keeping its current lease values). The button asks for a **second click** before doing that.
- **Remove multi-IPs before you leave.** Windows keeps them across reboots, and the app warns you if any are still active when you hide or quit.

4. PING — connectivity monitor

- Up to 8 hosts. **+ GATEWAY** and **+ MY IP** add the two most useful ones.
- Checks use a TCP connection to port 80, not ICMP, so they work through switches and appliances that drop ping.
- Dot colour: green under 150 ms, amber at 150 ms and above, red for no response. The sparkline is the last 60 samples.
- **TRCRT** runs a live traceroute to that host. **COPY** puts the whole summary on the clipboard for the service report.
- **PAUSE** freezes the display and keeps the data; **STOP** clears everything.

5. SCAN — subnet discovery

Sweeps a /24 with ICMP and ARP, probes ten common service ports on every responder, then resolves names.

- **BASE IP** is filled from the adapter. Narrow the range (for example 1-60) to finish faster.
- **Site picker** loads a saved site's subnet. Scanning a known subnet compares the results against that site's device list.
- Each row shows IP, response time, MAC, **hostname**, vendor, and open ports. Your own laptop is marked **SELF**.

- Hostnames come from reverse DNS or, failing that, a NetBIOS name query — which is what most Windows PCs, NAS units, printers and NVRs answer.
- Vendor names come from a bundled offline database of 57,000 manufacturer prefixes. **ONLINE VENDOR LOOKUP** (below the scan button) controls whether misses are looked up on macvendors.com; your IT department can force it off.
- **OPEN** launches the device's web interface. **RDP** and **SSH** do what they say. RTSP cameras without a web UI get their stream URL copied instead.
- **SAVE SITE** stores the results for next time.

> After scanning a known site's subnet an amber chip appears under the results: "SITE — +2 NEW · 1 MISSING". Press VIEW → to jump to the site with those changes listed. The chip stays until you dismiss it or start another scan; the app never switches tabs on its own.

6. SITES — site knowledge base

One record per site, one record per device, devices anchored by MAC address so an IP change is tracked instead of creating a duplicate.

- **Site card:** name, tag, last scan, device count, and change badges since the previous scan.
- **Device drawer:** click a device for MAC, hostname, first and last seen, open ports, notes, and credentials.
- **Type chip:** click to cycle CAMERA · NVR/DVR · NETWORK · SERVER · WORKSTATION · ACCESS CTL · PRINTER · OTHER. Types are guessed from the vendor and can be corrected.
- **SCAN NOW** loads the site's subnet into SCAN; the results come back as a change chip.

Credentials

Each device can store any number of label/value pairs (web UI login, ONVIF user, and so on). Values are masked; the eye icon reveals one at a time.

On disk, credential values are encrypted with Windows Data Protection (DPAPI). They can only be read by your Windows account on this machine. Records created by earlier versions are converted automatically the first time the app opens them.

Export and import

Button	What you get
EXCEL	A formatted workbook of every site and device — the right thing to send to a customer or another tech.
JSON	A portable file for moving your data to another machine. Credentials are left out unless you choose WITH CREDS, which writes them as readable text — only do that for a file you'll move and then delete.
IMPORT	Reads a JSON export. Your current data is backed up first (the last five backups are kept). MERGE keeps everything you have and adds what's new, matching devices by MAC; REPLACE discards everything and loads the file.

7. DIAGNOSTICS

Click the version chip in the title bar.

- **STATE** — version, whether the app is elevated, your user and machine, where the data lives, credential storage status, and any policy in effect.

- **LOG** — every adapter change the app made, with the exact commands run, their output, and whether the change was confirmed. Also app launches, backups, imports, and scan name-resolution timings. Survives restarts.
- **COPY DIAGNOSTICS** — puts a plain-text report on the clipboard.

> If something doesn't work, that report is what to send. It says exactly what Windows was asked to do and what Windows said back.

8. Common workflows

Arriving at a new site

Plug in. ETHER shows the adapter and its current state in LIVE. Pick a preset or a Known Default, APPLY, and the status line confirms the change. If the device you need is on a second subnet, add a Multi-IP instead of changing your primary.

Finding a camera that's stopped answering

SITES → the site → SCAN NOW. When the sweep finishes, the change chip tells you if it's MISSING (not on the wire), MOVED (new IP — usually DHCP), or present with a different open-port pattern (rebooted into a bad state). VIEW → shows which.

"It pings but the web page won't load"

ETHER → MTU. If the adapter was left at a jumbo or PPPoE-sized MTU by something else, one click fixes it.

Handing a site to another technician

SITES → EXCEL for a document they can read anywhere, or JSON if they also run NET//ETHER and should import the records.

9. Data and privacy

- All data is stored under %APPDATA%\net-ether on the technician's laptop.
- No software is installed on customer systems and no customer network data leaves the laptop.
- Stored credentials are encrypted per user and per machine.
- The only outbound internet request is the optional vendor lookup, and it sends only the first half of a MAC address.
- Uninstalling the application leaves the data folder in place.